

Fermi-surface sampling on the thermal sunrise

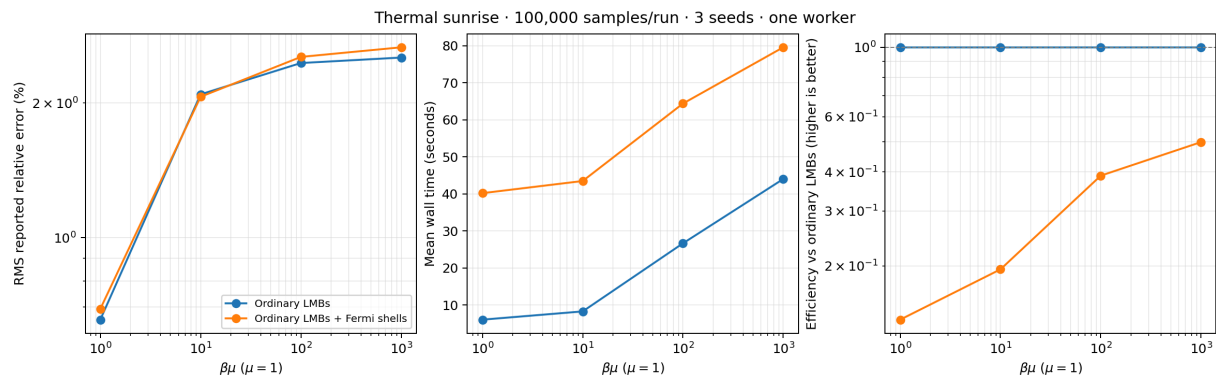
Measured result

Adding the independent Fermi-pair channel to ordinary optimized LMB sampling does not improve this sunrise benchmark, including at $\beta = 1000$. Its integration uncertainty is nearly unchanged while the map adds substantial cost.

Measured on 3 October 2026 (Europe/Zurich), at revision 2a236eaab703fe7e9b5505522bf6884efa4b9ef7, using the Nix ci-optim binary (GammaLoop optimization level 2, dependency optimization level 3, no debug information) and assembly-compiled integrand evaluators. Each entry uses three seeds, each with 100,000 samples in ten iterations. Error percentages below are RMS reported errors of individual runs, relative to the analytic target; times are arithmetic means. Generation took 3.22 seconds, shared by every run.

β	Ordinary error	+ Fermi error	Ordinary time	+ Fermi time	Efficiency
1	0.655%	0.693%	6.05 s	40.18 s	0.135
10	2.092%	2.066%	8.27 s	43.46 s	0.195
100	2.458%	2.537%	26.65 s	64.37 s	0.389
1000	2.529%	2.666%	44.00 s	79.54 s	0.498

Efficiency is the ordinary catalogue's mean $\sigma^2 t$ divided by the Fermi catalogue's mean $\sigma^2 t$; larger is better. The corresponding estimated equal-error time penalties are 7.42, 5.13, 2.57 and 2.01. They describe convergence at this sample budget, not an independently timed run to a fixed target precision.



All 24 primary-comparison runs agree with the analytic value within 1.61 reported standard errors. All eight three-seed means agree within 1.57 standard errors. The real component and its uncertainty are exactly zero. Across all 60 runs (six million draws, including diagnostic modes), no NaN or unstable sample was rejected. Fermi channels received 16.2%, 31.0%, 32.4% and 32.3% of draws as beta increased; the map was actively exercised.

The average mapping cost rises from 26–33 microseconds per draw to 354–390 microseconds. The colder runs have more arbitrary-precision evaluation under the unchanged stability policy: at $\beta = 1000$, the ordinary and augmented catalogues use Arb for 21.5% and 19.8% of samples, respectively. This makes the relative map overhead smaller at high beta; it does not produce a variance gain. CPU times are within 1% of wall times, and matching preliminary sequential runs reproduce identical numerical values with timings within about 2% of these concurrent batches.

A more expensive integrand can amortize the roughly 330–360 microsecond extra mapping cost, but a useful speedup also requires variance reduction. This sunrise shows no substantial reduction. Its

only nominal improvement is a 2.5% variance reduction at $\beta = 10$; holding that noisy estimate fixed would require about 13.8 milliseconds of extra common work per draw to break even. This is a cost-model extrapolation, not evidence for a more complicated integrand.

The separate three-loop follow-up measures the user's tennis-ball and Mercedes examples, including a targeted single-shell channel that improves cold tennis-ball efficiency.

Convergence diagnostic

The single-LMB, Fermi-only and same-parent mixed modes omit explicit ordinary sampling of the soft gluon momentum. At $\beta = 10$, individual single-LMB and Fermi-only runs miss the reference by 5.31 and 4.90 reported errors, respectively; the three-seed single-LMB and mixed estimates miss by 4.47 and 3.82 errors. Their reported errors also vary greatly across seeds. These failures demonstrate that a small nominal error from a single such run is insufficient validation. The optimized catalogue includes gluon-momentum LMBs and passes the independent reference checks, including a separate 200,000-sample control. This behavior is consistent with undersampling the soft-gluon region in the single-parent maps. Keep ordinary soft-momentum coverage when adding Fermi channels.

The complete measurements are in results.json all-mode metrics are in summary.csv and summary.json. The independent control is in validation-results.json. Per-run command cards, logs and resumable workspaces remain under /tmp/fermi-sunrise-benchmark/scan/. No production implementation was changed.

Scope and reference

This benchmark compares changes of variables for the physical massless quark-gluon sunrise, using tests/resources/graphs/sunrise_qcd_vacuum.dot. It tests correctness, variance reduction and sampling overhead on a simple two-loop integral. It does not measure the performance of more complicated integrands.

The model is SU(3), with one massless quark, $a_5 = 1$, $\mu_B = 3$, hence quark chemical potential $\mu = 1$ and Fermi radius $p_F = 1$. Generation includes vacuum subtraction and local plus integrated UV counterterms, with $m_{uv} = \mu_r = \text{renormalization_localization_scale} = 2$. Threshold subtraction is disabled. All physical orientations are summed.

For inverse temperature β , the imaginary integral is compared with

$$I_{\text{ref}} = -\left(5\frac{\pi}{18\beta^4} + \frac{1}{\pi\beta^2} + \frac{1}{2\pi^3}\right).$$

This is the one-flavor part of the order- g^2 pressure in Vuorinen, Eq. (3.15), with $g^2 = 4\pi$, omitting the pure-gluon contribution. The real component must vanish.

Controlled comparison

ordinary uses the two fermion momenta as its ordinary spherical LMB. fermi replaces both radial maps with the product `product(block(lmb(0), fermi(0)), block(lmb(1), fermi(1)))`. mixed samples one of these two channels per draw. All use `map_density` partitioning, `power = 2`, `b = e_cm = 1`, and the same physics and stability settings. The primary comparison is `optimized` versus `optimized_fermi`: automatic optimized LMB coverage, respectively without and with the additional Fermi channel. The ordinary catalogue contains (1,2), (0,2) and (0,1); the first two explicitly sample the massless gluon momentum. The single-parent modes serve as convergence diagnostics.

Both pipelines share model loading, generation, evaluator compilation, stability checks, thermal physics, subtraction, and accumulation. The first difference is the sampling catalogue and its map/partition; subsequent adaptive grids train independently. Every run starts a fresh integration

workspace. Jobs are shuffled deterministically; one integration worker is used throughout. The recorded scan runs one sequential batch per temperature on CPUs 96, 97, 98 and 99, respectively, with the four temperature batches concurrent. The host is a shared AMD EPYC 9754 machine. Matching preliminary sequential runs provide a check on timing repeatability.

For each mode and temperature, the analysis reports RMS integration uncertainty over seeds and the arithmetic mean runtime. The combined estimate is the unweighted mean over independent seeds, with error $\frac{\sqrt{\sum_i \sigma_i^2}}{n}$. Efficiency is the optimized ordinary catalogue's mean $\sigma^2 t$ divided by the candidate's mean $\sigma^2 t$; values above one favor the candidate. Both the single-LMB and optimized-LMB comparison columns are retained in the machine-readable summary. This is an estimate of equal-error time efficiency at the measured sample budget, not a measured long-run speedup. Startup, loading, warm-up and result persistence are included in run wall time. Generation is timed separately. CPU time, internal mapping/evaluation time, precision escalation and rejected sample fractions are retained in the raw results.

When variance improves by a factor $G > 1$, an extra common evaluation cost c per sample would give the approximate efficiency $G \frac{t_o + c}{t_f + c}$, where t_o and t_f are measured per-sample costs. The break-even extra cost is $\max\left(0, \frac{t_f - G t_o}{G - 1}\right)$. This holds the measured variance and adaptation behavior fixed; it does not establish a gain for a different physical integrand.

Reproduction

Run from the repository root inside `nix develop`, with a binary built from the measured revision and its configured Symbolica runtime license:

```
python examples/cli/eos/fermi_surface_sampling/benchmark.py \
  --binary /path/to/gammloop --output /tmp/fermi-sunrise \
  --generate --cpu 96
python examples/cli/eos/fermi_surface_sampling/summarize.py \
  /tmp/fermi-sunrise/results.json --output /tmp/fermi-sunrise/summary
```

The driver uses only the Python 3.11+ standard library; plotting additionally requires Matplotlib. Each run retains its overlay, command card, full log, integration workspace and exact result JSON. `results.json` records source revision, binary identity and generation-card digest. The default scan is $\beta = 1, 10, 100, 1000$, three seeds (1337, 2843, 4517), five sampling modes, and 100,000 draws in ten equal iterations.